A STRATEGY FOR THE DROP

Status of the claims. Every claim is labelled. *Proved*: a complete argument from the stated inputs; when these include the exact triangles of §2, we say so. *Computed*: an exact machine computation, checked by an independent program unless said otherwise. *Plausible*: an argument with a named gap. *Conjecture*: expected, with no argument. The only facts about energy used anywhere are $E \geq 0$, and $E > 0$ for the instantons counted by a map to which no flat connection can contribute.

Let $K \subset S^3$ be a knot with $\Delta_K = 1$, let $Y_N = S^3_N(K)$, and let P_0, P_1 be the SO(3)-bundles on $Y_{\pm 2}$ with $w_2 = 0$ and $w_2 \neq 0$. A cosmetic pair $Y_2 \cong Y_{-2}$ is excluded by a *drop* of consistent sign: either $D = \min(\lambda_0, \lambda_1)(Y_2) - \min(\lambda_0, \lambda_1)(Y_{-2}) > 0$, or $m_1 = \lambda_0(Y_2) - \lambda_1(Y_{-2}) > 0$ and $m_2 = \lambda_1(Y_2) - \lambda_0(Y_{-2}) > 0$ (λ_i as in §2). We explain what was wrong with the previous note (rho.pdf), prove that the formal Floer-theoretic inputs cannot give any drop, record what they do give, and propose a replacement: a statement about where the flat connections sit, read off one curve in the pillowcase, which needs neither energy nor differentials.

1. What was wrong with rho.pdf

Step 2 assumed that the differentials vanish. Its dichotomy "$a(x) = 0$, or $a(x) \neq 0$ in $T(K)$" omits the case that $a(x)$ is a nonzero boundary. Then Lemma 2.2 gives only $\Lambda(Y_{-2}) \leq \max\big(\Lambda(Y_2) - \frac{1}{8} - E,\, R(ax) - E'\big)$, where $R(ax)$ is the least level of a primitive of $a(x)$, and this may exceed $\Lambda(Y_2)$. Computed: a triangle obeying every identity and every level of Table 1, with all energies positive and with *acyclic* middle term (one finite bar of Chern–Simons length 37/200), has drop $-1/50$.

Step 4 was an energy estimate equivalent to the conclusion. It asserted that a costs energy more than $1/16$ on the lowest class. In a cosmetic pair the triangles force the lowest classes of both bundles to lie within $1/16$ above the lowest traceless class, for perfect complexes (Proposition 4.2). Step 4 is the negation of this, so it assumes that there is no cosmetic pair. Positivity of energy cannot replace it: the model of §3 realises every cost in $(0, \frac{1}{16}]$.

The level c is not a filtration (proved). By (2.1) with $i_0 = 1$, an index-one trajectory of energy E from α to β has $c(\beta) - c(\alpha) = \frac{1}{8} - E$: the differential lowers $\widetilde{\mathrm{CS}}$ but *raises* c whenever $E < \frac{1}{8}$. So levels are meaningful only for homogeneous chains; λ_i and DLME's injectivity lemma are unaffected, but boundary depth $\beta(C)$ must be measured in $\widetilde{\mathrm{CS}}$ within one degree. A boundary z then has a primitive of level at most $c(z) + \beta(C) - \frac{1}{8}$, since the primitive has degree one higher; the earlier bound without the $-\frac{1}{8}$ loses exactly the margin that DLME's argument needs.

Two smaller points. Step 3 needs that no irreducible traceless representation sends the longitude to ± 1; $\det K = 1$ excludes only the binary dihedral ones, so this is an extra hypothesis. And "the differential vanishes for every Seifert fibred Y" was checked, by parity, only for the computed examples.

2. The setting

Levels. Work over $\mathbb{F}_2$ with DLME's $\mathbb{Z}$-graded lifts, filtered by $\widetilde{\mathrm{CS}}$ in each degree. A generator has level $c(\alpha) = \widetilde{\mathrm{CS}} - \widetilde{\mathrm{gr}}/8 = (3 + \rho(\alpha))/16$, with ρ the adjoint ρ-invariant (proved: rho.pdf, Proposition 2.1), in this normalisation for both bundles. A homogeneous chain has the largest level of its generators, a homogeneous class the least level of a representing cycle, and $\ell(C)$ is the least level of a nonzero homogeneous class of $H(C)$. Then $\lambda_i(Y) = \ell(C(Y, P_i))$ (so λ_0 is DLME's ℓ), $\lambda_i^+(Y)$ is the largest level at which an infinite bar is born, $\Lambda = \min(\lambda_0, \lambda_1)$, and μ, μ^+ are the same invariants of the traceless complex. Reversing orientation sends c to $\frac{3}{8} - c$ (proved: ρ changes sign).

Level formula (proved: Atiyah–Patodi–Singer). If an instanton of energy E on a cobordism W, counted over a family of dimension g with insertions of degree k, runs from α to β, then, with $i_0 = k - g$,

$$c(\beta) - c(\alpha) = \frac{i_0 + 3b^+(W)}{8} - E - \sum_j \frac{3 - h(\gamma_j) + \rho(\gamma_j)}{16}, \tag{2.1}$$

summed over the middle ends with limits γ_j (singular maps along a surface F carry an extra $-(\chi(F) + F \cdot F/2)/8$, which is 0 for all surfaces below). The *level* $\mathrm{lev}(f)$ of a map is the supremum of the right side over counted instantons. On $L(4,1)$ the flat connection γ_1 (rotation by $\pi/2$, $w_2 \neq 0$) contributes $\frac{3}{16}$ or $\frac{1}{16}$ according to orientation, so the maps carrying a sixteenth exchange P_0 and P_1.

The triangles (assumed; their construction is plausible, and is not the difficulty). Let $W' = (-X_2^\circ) \cup_{S^3} X_{-2}^\circ \colon Y_2 \to Y_{-2}$, and $S \subset W'$ the sphere of square -4 formed by the cores, with $\partial \nu S = L(4,1)$. Let $C^{\mathrm{tl}}(K)$ be the irreducible complex of Kronheimer–Mrowka's singular theory of (S^3, K) with traceless meridian. We use

$$\text{(I)}\quad C(Y_2, P_0) \xrightarrow{a} C^{\mathrm{tl}} \xrightarrow{b} C(Y_{-2}, P_1) \xrightarrow{c} C(Y_2, P_0), \qquad \text{(II)}\quad C(Y_2, P_1) \xrightarrow{a'} C^{\mathrm{tl}} \xrightarrow{b'} C(Y_{-2}, P_0) \xrightarrow{c'} C(Y_2, P_1),$$

with homotopies g_A (from Y_2 to Y_{-2}), g_M, g_B satisfying $dg + gd = f'f$ for consecutive maps f, f', and $fg + gf \simeq \mathrm{id}$ at

each vertex (DLME's triangle detection). Their sum is the triangle obtained invariantly on the double cover branched along S. The two lifts of the involution to the bundle give P_0 and P_1, so the outer terms are $A = \bigoplus_i C(Y_2, P_i)$ and $B = \bigoplus_i C(Y_{-2}, P_i)$. The two isotropy types along the fixed knot (trivial, rotation by π) give the middle term M, a copy of $C(S^3) \oplus C^{\mathrm{tl}}(K)$ for each triangle; $C(S^3) = 0$ in the irreducible theory, and it enters only by excluding breaking at the trivial connection. The levels, from (2.1), are:

map	cobordism	terms of (2.1)	level
a, a', b, b'	$(-X_2^\circ, D_l)$, (X_{-2}°, D_r), singular along the cores	none	$-E$
c, c'	$-(W' \setminus \nu S)$, limit γ_1	$b^+ = 1$; γ_1: $\frac{3}{16}$	$\frac{3}{16} - E$
g_A, g'_A	W', interval of metrics	$i_0 = -1$	$-\frac{1}{8} - E$, with $E > 0$
g_M, g_B	composites through $-(W' \setminus \nu S)$, interval	$b^+ = 1$, $i_0 = -1$; $\frac{3}{16}$	$\frac{1}{16} - E$
pentagon maps	the same, two parameters	$b^+ = 1$, $i_0 = -2$; $\frac{3}{16}$	$-\frac{1}{16} - E$

TABLE 1. Levels (proved, as bookkeeping).

For g_A, $E > 0$ because W' is simply connected and a flat singular contribution would need the longitude to go to -1 and to $+1$ at once. For a, b' (resp. a', b), $E = 0$ only at a flat singular connection, i.e. a traceless representation sending the longitude to -1 (resp. $+1$).

Filtered algebra (proved; arbitrary differentials). For a boundary z let $R(z)$ be the least level of a primitive; by §1, $R(z) \le c(z) + \beta(C) - \frac{1}{8}$.

Lemma 2.1 (DLME)**.** *If a chain map $f\colon C \to C'$ is injective on homology, then $\ell(C') \le \ell(C) + \mathrm{lev}(f)$.*

Lemma 2.2. *Let $C \xrightarrow{f} C' \xrightarrow{f'} C'' \xrightarrow{f''} C$ come with triangle-detection data, $g\colon C \to C''$, $g'\colon C' \to C$. Then $(f, g)\colon C \to \mathrm{Cone}(f')$ is injective on homology. If x is a homogeneous cycle with $[x] \neq 0$, then either $[fx] \neq 0$, a class of level at most $c(x) + \mathrm{lev}(f)$; or $fx = dy$, and for every such y the cycle $gx + f'y$ is not a boundary, so $\ell(C'') \le \max\big(c(x) + \mathrm{lev}(g),\, R(fx) + \mathrm{lev}(f')\big)$. If $fx = 0$, then $\ell(C'') \le c(x) + \mathrm{lev}(g)$.*

Proof. $\psi(m, z) = g'm + f''z$ is a chain map $\mathrm{Cone}(f') \to C$ with $\psi \circ (f, g) = g'f + f''g \simeq \mathrm{id}$. In the cone, $(fx, gx) - d(y, 0) = (0, gx + f'y)$ represents the nonzero class of x. □

The term $R(fx)$ cannot be dropped (the example of §1).

3. The no-go theorem

Theorem 3.1 (proved, by explicit models verified by five independent programs)**.** *There are finite filtered complexes over $\mathbb{F}_2$ for $(Y_{\pm 2}, P_i)$ and for the middle term, with maps and homotopies, such that: (a) both triangles satisfy triangle detection, hence are exact, and every level of Table 1 holds, strictly where energy is positive, and even with a, a', b, b' strictly negative, so that no flat connection contributes anywhere; (b) the differentials have positive energy, and may be zero or nonzero; (c) the maps have the actual degrees mod 4; the twist by the character of $H_1(Y_{\pm 2}) = \mathbb{Z}/2$ acts freely with its actual degrees and commutes with everything; Euler characteristics vanish, ranks are equal in all degrees, the traceless homology is nonzero, and $b'a = 0 = ba'$; (d) the data are dual to those of the mirror knot, carry the odd-degree automorphisms that a cosmetic pair would provide, and lift to the branched double cover with DLME's levels; (e) the complexes of Y_2 and Y_{-2} coincide bundle by bundle. In all of them $\lambda_i(Y_2) = \lambda_i(Y_{-2})$ and $\lambda_i^+(Y_2) = \lambda_i^+(Y_{-2})$, so every drop, at the bottom and at the top, is 0. Hence no argument whose inputs are (a)–(e), the two energy facts, and filtered algebra with arbitrary differentials can prove $D > 0$, $m_1 > 0$ or $m_2 > 0$.*

The simplest model has zero differentials. Each bundle of Y_2 and of Y_{-2} has one generator, at level 0; in each triangle the middle term has generators y at level $-\varepsilon$ and y' at level t, with $0 < \varepsilon, t \le \frac{1}{16}$. The map a sends the generator of Y_2 to y, b sends y' to the generator of Y_{-2}, g_M and g_B invert these, and $c = g_A = 0$. Variants with nonzero differentials send the lowest class of Y_2 to a boundary in the middle complex; doubling and tensoring give (c) and (d).

What escapes the theorem. (i) The *positions* of the generators, that is, their ρ-values: in the models the levels are free parameters, while in Floer theory they are values of one function along one curve (§5). (ii) Which generators survive in homology. (iii) Structure not listed: the action of the point class, the reducibles, triangles through other slopes. (Through Y_0 nothing is gained: the composites are families on W' whose levels are already in Table 1; plausible.) (iv) Energy estimates, which are excluded.

4. What the Floer layer does give

For a filtered complex X and $t \in \mathbb{R}$, let $N_X(t)$ be the number of infinite bars born at level $\le t$, and $P_X(t)$ the number of bars, finite or infinite, alive at level t; both are summed over one period of degrees, a degree-d chain having level $\le t$ when its $\widetilde{\mathrm{CS}}$ is $\le t + d/8$, so that everything below is homology of genuine subcomplexes. Put $\bar{P}_X(t) = \sup_{s \le t} P_X(s)$.

Proposition 4.1 (proved, given the triangles; arbitrary differentials)**.** $N_B(t-\frac{1}{8}) \geq N_A(t) - P_M(t)$ *for every* t*. Hence, with* $t_* = \inf\{t : N_A(t) > \bar{P}_M(t)\}$*,*

$$\Lambda(Y_{-2}) \leq t_* - \tfrac{1}{8}, \qquad D \geq \tfrac{1}{8} - \big(t_* - \Lambda(Y_2)\big).$$

Proof. $\Phi = (a, g_A)\colon A \to \mathrm{Cone}(b)$ is injective on homology (Lemma 2.2). Filter the cone so that $m \in M$ sits at level $c(m) - \frac{1}{8}$; this is a filtered complex since $\mathrm{lev}(b) \leq 0$, and Φ lowers levels by at least $\frac{1}{8}$. Let V be a lift to the filtered homology at level t of the classes of A born by t, and U the kernel of $x \mapsto [ax]$ from V to the filtered homology of M at level t; $\dim U \geq N_A(t) - P_M(t)$. By the filtered exact sequence $0 \to B \to \mathrm{Cone}(b) \to M[1] \to 0$, $\Phi_*(U)$ comes from classes of B born by $t - \frac{1}{8}$, and these are independent because Φ_* is injective. □

Computed: $t_* - \Lambda(Y_2) \leq \frac{1}{32}$ for the 80 torus knots with $pq \leq 60$, and ≤ 0.0294 for the 54 cases with $\det K = 1$, $pq \leq 120$. But the hypothesis $t_* < \Lambda(Y_2) + \frac{1}{8}$ fails for the figure-eight knot, whose drop is $\frac{1}{8}$, and it never holds in the $\pm 1/n$ analogue (§6). It needs a lower bound on the number of *homology classes* of Y_2 near its bottom, which nothing available supplies. In a cosmetic pair $A \cong B$, so $N_Y(t - \frac{1}{8}) \geq N_Y(t) - P_M(t)$ for all t; for perfect complexes this says that spectral values of Y that are r apart differ by more than $\frac{1}{8}$, where r is the rank of the traceless homology modulo the twist (proved). Every computed $Y_{\pm 2}$ with $\det K = 1$ has $r + 1$ consecutive spectral values within 0.013 (computed).

Proposition 4.2 (proved, given the triangles for K and its mirror)**.** *Suppose* $Y_2 \cong Y_{-2} = Y$ *and no irreducible traceless representation sends the longitude to* ± 1*. If the complexes are perfect, then*

$$\mu < \Lambda(Y) \leq \max(\lambda_0, \lambda_1)(Y) \leq \mu + \tfrac{1}{16}, \qquad \mu^+ - \tfrac{1}{16} \leq \min(\lambda_0^+, \lambda_1^+)(Y) \leq \max(\lambda_0^+, \lambda_1^+)(Y) < \mu^+.$$

In general either this holds with $\frac{1}{16}$ *replaced by* $\frac{1}{16} + \beta(Y)$*, or the traceless complex has a finite bar of* $\widetilde{\mathrm{CS}}$*-length at least* $\frac{1}{8}$ *alive at level* $\Lambda(Y)$*.*

Sketch. Let x realise Λ, say in P_0. If $[ax] = 0$, Lemma 2.2 at the vertex Y_2 of (I) puts a class of Y_{-2} below Λ; so $[ax] \neq 0$ has level below Λ, and $\mu < \Lambda$. Every traceless class below Λ is killed by b and b' (otherwise they carry it below Λ); Lemma 2.2 at the middle vertex of (I) and of (II), applied to the lowest one, bounds λ_0 and λ_1 by $\mu + \frac{1}{16}$. The top is the same statement for the mirror, since $S^3_r(\bar{K}) = -S^3_{-r}(K)$. □

These windows are necessary, not sufficient: the mirrors of torus knots with det = 1 satisfy the bottom one (for $T(3,-5)$, $\Lambda(Y_2) = -0.6255$ and $\mu = -0.6292$) and still have $D \approx \frac{1}{8}$ (computed).

Why $1/16$ *is a ceiling, not a floor.* The constant enters only through c, c', g_M, g_B, which move classes upward: from Y_{-2} or the middle term to Y_2, or from Y_{-2} to the middle term. Through Lemma 2.2 it bounds Λ above by $\mu + \frac{1}{16}$, never a drop below. The two maps from the Y_2 side to the Y_{-2} side of level $-\frac{1}{16}$ do not help. The exterior $W' \setminus \nu S$ with limit γ_1 is the $L(4,1)$-face of the one-parameter odd family on W', whose S^3-face is empty, so it is null-homotopic (plausible: the gluing at γ_1 is not written). The composite $g_A g_M$ plus pentagon terms is a chain map $C^{\mathrm{tl}} \to C(Y_{-2}, P_1)$ of level $-\frac{1}{16}$ that vanishes in the model of §3 (proved). Every lower bound comes from the $-\frac{1}{8}$ of g_A, and from counting.

5. The pillowcase reduction

The curve. Let $\mathcal{C}$ be the image of the irreducible $\mathrm{SU}(2)$ representations of $\pi_1(S^3 \setminus K)$ in the pillowcase, with meridian $\sim e^{2\pi i\alpha}$, longitude $\sim e^{2\pi i\beta}$, $\delta = \alpha - \frac{1}{4}$, and slope $s = d\beta/d\alpha$. Its closure meets the line $\beta = 0$ of reducibles only where $e^{4\pi i\alpha}$ is a root of Δ_K (proved: Burde, de Rham), so for $\Delta_K = 1$ it is a union of closed curves, which we take to be smooth and immersed (plausible after a small holonomy perturbation). The irreducible flat connections on (Y_N, P_0) and (Y_N, P_1) are the points of $\mathcal{C}$ on $N\alpha + \beta \in \mathbb{Z}$ and on $N\alpha + \beta \in \frac{1}{2} + \mathbb{Z}$, each $\mathrm{SO}(3)$ connection appearing twice, as x and its twist by $\mu \mapsto -\mu$; the traceless generators are the points on $\alpha = \frac{1}{4}$.

The formula for c *(plausible).* There is a function Φ on $\mathcal{C}$ with

$$d\Phi = 2\big(\alpha - \tfrac{1}{4}\big)\, d\beta,$$

such that every flat connection x on Y_N, in either bundle, has

$$c(Y_N, x) = \Phi(x) + \tfrac{3}{16} \operatorname{sgn} N - N\alpha\big(\tfrac{1}{2} - \alpha\big) - \tfrac{1}{8}\iota_N(x), \qquad \iota_N(x) \in \{0, 1\}, \tag{5.1}$$

where $\iota_2(x) = 0$ if $s(x) \notin (-2, 0)$ and $\iota_{-2}(x) = 0$ if $s(x) \notin (0, 2)$; a traceless generator z has level $\Phi(z) + \operatorname{sgn} s(z)/16$. As $\alpha(\frac{1}{2} - \alpha) = \frac{1}{16} - \delta^2$, for $N = \pm 2$ this reads

$$c(Y_{\pm 2}, x) = \Phi(x) \pm \big(\tfrac{1}{16} + 2\delta^2\big) - \tfrac{1}{8}\iota_{\pm 2}(x). \tag{5.2}$$

Status: computed, exactly for torus knots (all slopes, both bundles) and to within 10^{-4} for the figure-eight knot ($N = \pm 1, \pm 2, \pm 3$); on every arc of 16 mirror torus knots, Φ recovered from Y_2 and from Y_{-2}, in both bundles, is the same rational number. In general it is plausible (we estimate 70%, and 55% for the rule for ι): it is the $\mathfrak{su}(2)$ analogue of the splitting theorem of Boden–Herald–Kirk–Klassen, namely a Kirk–Lesch splitting of ρ along the boundary

torus, with ι a Maslov triple index and Kirk–Klassen's formula for the Chern–Simons part; degenerate curves need perturbations and a limit.

How 1/16 *enters.* At $\alpha = \frac{1}{4}$ the slope term of (5.1) is $\pm\big(\frac{3}{16} - |N| \cdot (\frac{1}{4})^2\big) = \pm\frac{3-|N|}{16}$, since $\alpha(\frac{1}{2} - \alpha)$ attains its maximum $(\frac{1}{4})^2$ exactly at the traceless holonomy. The $\frac{3}{16}\operatorname{sgn} N$ comes from the signature of the trace; $-N\alpha(\frac{1}{2} - \alpha)$ is the solid-torus Chern–Simons term $N\alpha^2$ plus the solid-torus spectral flow $-N\alpha/2$, which are $\frac{2}{16}$ and $-\frac{4}{16}$ at $N = 2$, $\alpha = \frac{1}{4}$. At $|N| = 1$ the offset is $\frac{1}{8}$, the value of DLME's constant; at $|N| = 2$ it is $\frac{1}{16} = \frac{3}{16} - 2(\frac{1}{4})^2$. So, where ι vanishes, Y_2 lies at least $\frac{1}{16}$ above Φ and Y_{-2} at least $\frac{1}{16}$ below it, with equality at $\alpha = \frac{1}{4}$. The drop below is this gap $2 \cdot \frac{1}{16}$, less a spacing error.

Lemma 5.1 (proved, given (5.1))**.** *Let $\Delta_K = 1$, let $\mathcal{C}$ be as above, and $\Phi_0 = \min_{\mathcal{C}} \Phi$.*

(i) *The critical points of Φ on $\mathcal{C}$ are the traceless points and the horizontal tangencies. Near a traceless point z of slope s, $\Phi = \Phi(z) + s\delta^2 + O(\delta^3)$; so positive-slope traceless points are local minima and negative-slope ones local maxima; Φ_0 is attained at a positive-slope traceless point or at a horizontal tangency.*

(ii) *If $s \notin (-2, 0)$ at every flat connection of Y_2, then every flat connection of Y_2, in either bundle, has level $\geq \Phi_0 + \frac{1}{16}$; hence $\Lambda(Y_2) \geq \Phi_0 + \frac{1}{16}$, whatever the differentials.*

(iii) *Every traceless generator has level $\geq \Phi_0 - \frac{1}{16}$; if Φ_0 is attained at a traceless point, the lowest one has level $\leq \Phi_0 + \frac{1}{16}$.*

(iv) *Suppose $\Phi_0 = \Phi(z_0)$ with z_0 traceless, and that the branch of $\mathcal{C}$ through z_0 is a graph over $v = \beta - 2\alpha$ on $|v - v(z_0)| \leq \frac{1}{4}$, with slope $\geq s_* > 2$ there. Then some flat connection of Y_{-2} has level $\leq \Phi_0 - \frac{1}{16} + \frac{1}{16(s_*-2)}$. On $|v - v(z_0)| \leq \frac{1}{2}$ the same holds in each bundle, with $\frac{1}{4(s_*-2)}$ in place of $\frac{1}{16(s_*-2)}$.*

(v) *Let $G(Y)$ be the least level of an irreducible flat* SO(3) *connection on Y, over both bundles. Under (ii) and (iv), $G(Y_2) - G(Y_{-2}) \geq \frac{1}{8} - \frac{1}{16(s_*-2)}$, which is positive for $s_* > \frac{5}{2}$ and at least $\frac{1}{16}$ for $s_* \geq 3$; the bundle-by-bundle and mixed differences are at least $\frac{1}{8} - \frac{1}{4(s_*-2)}$.*

Proof. (i) Along $\mathcal{C}$, $d\Phi = 2\delta\, d\beta$, and near z, $d\beta = (s + O(\delta))\, d\delta$. (ii) By (5.2) with $\iota_2 = 0$, $c \geq \Phi + \frac{1}{16} \geq \Phi_0 + \frac{1}{16}$; and the level of a class is at least the least level of a generator. (iii) From the traceless rule and (i). (iv) The flat connections of Y_{-2} are the points of $\mathcal{C}$ with $v \in \frac{1}{2}\mathbb{Z}$ (both bundles; $v \in \mathbb{Z}$, $v \in \frac{1}{2} + \mathbb{Z}$ for each), so the interval contains one, x. On the branch $dv = (s-2)\, d\alpha$, so δ has the sign of $v - v_0$ and $|\delta| \leq |v - v_0|/(s_* - 2)$; and $d(\Phi - 2\delta^2) = 2\delta\, dv$. Hence $\Phi(x) - 2\delta(x)^2 - \Phi_0 = \int 2\delta\, dv \leq (v - v_0)^2/(s_* - 2)$, and (5.2) with $\iota_{-2} \geq 0$ gives the claim. (v) Subtract. □

Proposition 5.2 (proved)**.** *G is an invariant of the oriented manifold, and so are the multisets of levels of flat connections in each bundle. Hence a cosmetic pair has $G(Y_2) = G(Y_{-2})$.*

Proof. An orientation-preserving diffeomorphism pulls irreducible flat SO(3) connections back to irreducible ones, preserves ρ, which is intrinsic, and preserves w_2, since $H^2(Y_{\pm 2}; \mathbb{Z}/2) = \mathbb{Z}/2$. □

Corollary 5.3 (proved, given (5.1))**.** *If $\Delta_K = 1$ and $\mathcal{C}$ satisfies the hypotheses of (ii) and (iv) with $s_* > \frac{5}{2}$, then $S^3_2(K) \ncong S^3_{-2}(K)$. No energy enters, not even $E \geq 0$, and differentials play no role.*

Parts (ii) and (iii) explain §3: when they apply, the traceless bottom lies at or below the generator bottom of Y_2, so the middle term is always placed to receive the lowest class, and the triangles cannot decide. Conversely, by (v), the model of §3 cannot come from a curve satisfying (ii) and (iv).

The remaining gap. (a) *Shallow parts of $\mathcal{C}$.* If Φ_0 is attained at a horizontal tangency, or a flat connection of Y_2 has slope in $(-2, 0)$, then ι_2 may be 1 there and that generator may sit near $\Phi_0 - \frac{1}{16}$; the lemma then says nothing. Nothing known forces $|s| > 2$ on $\mathcal{C}$ for a general knot. (b) *The formula* (5.1), plausible only. (c) A drop of Λ itself, rather than of G, would also need a class of Y_{-2} below $\Phi_0 + \frac{1}{16}$, for instance that the lowest flat connection of Y_{-2} on the branch through z_0 survives in homology. This concerns differentials, not energy, and it is open; the cosmetic application does not need it.

6. **Data**

The ±2 surgeries (computed; Seifert fibred, zero differentials). For positive $T(p, q)$, $D > \frac{1}{8}$ and $m_1, m_2 \in (\frac{1}{8}, \frac{3}{16}]$, while $\lambda_0(Y_2) - \lambda_0(Y_{-2}) = -\frac{1}{8} + \frac{2n-5}{n^2-4}$ ($n = pq$) changes sign. For negative $T(p, q)$ ($pq \leq 120$), $D = \frac{1}{8}$ exactly when $\det > 1$ (78 knots: both lowest generators are one representation shared by $Y_{\pm 2}$, and $\frac{1}{8}$ is the level of a flat connection on $W' \setminus \nu S$), and $D = \frac{1}{8} - \frac{1}{4(n^2-4)}$ when $\det = 1$ (27 knots), least $\frac{1}{8} - \frac{1}{884}$ at $T(3, -5)$. For the figure-eight knot, $D = \frac{1}{8}$, $m_1 = \frac{13}{80}$, $m_2 = \frac{1}{16}$; in both cases with $m_2 = \frac{1}{16}$ (the figure-eight knot, $T(2, -3)$) the lowest class of Y_2 maps at zero energy through a binary dihedral representation, which does not exist when $\det K = 1$. At all 1732 shared representations the gap is $\frac{1}{8}$ (same bundle) or $\frac{3}{16}$ (mixed). Lemma 5.1 holds on 16 mirror torus knots and the figure-eight knot; (iv) is attained exactly for odd–odd mirrors, whose minimum of Φ sits midway between flat connections of Y_{-2}, and $m_2(T(2, -3)) = \frac{1}{16} = \frac{1}{8} - \frac{1}{4(6-2)}$ is sharp. For the figure-eight knot, Φ has one minimum,

$\frac{11}{80}$, at the traceless passage of slope $2\sqrt{5}$, and one maximum, $\frac{19}{80}$; $G(Y_2) = \frac{1}{5} = \Phi_0 + \frac{1}{16}$, $G(Y_{-2}) = \frac{3}{40} \le 0.1003$. The lemma does not apply to positive torus knots, whose arcs end on the reducible line with Φ concave; their drop exceeds $\frac{1}{8}$ by parity.

The $\pm 1/n$ surgeries (computed; torus knots). There is no positive one-step bound: for positive $T(p,q)$, $\ell(Y_{1/(n+1)}) - \ell(Y_{1/n}) = \frac{1}{4(pqn-1)(pq(n+1)-1)} \to 0$, and on the negative side the least one-step drop is $\frac{7}{232}$ ($T(3,-5)$, $n=1$). These are differences of Chern–Simons values across simply connected negative definite cobordisms, where only $E > 0$ is available. The *two-step* drop satisfies $\ell(Y_{-1/n}) - \ell(Y_{-1/(n+2)}) \ge \frac{1}{8}$ on all torus knots with $pq \le 160$, both chiralities, $n \le 3$, and on 18 knots with $n \le 8$; equality holds exactly for mirrors of odd–odd torus knots at odd n. The equality comes from flat SO(3) connections with traceless meridian and longitude (plausible as bookkeeping; computed). Such an x sits at $(\frac{1}{4}, \pm\frac{1}{4})$; for odd n, one of x and its twist by $\mu \mapsto -\mu$ is a flat connection of $Y_{-1/n}$ and the other of $Y_{-1/(n+2)}$, and $\operatorname{ad} x$, common to both, descends to the complement of the sphere of square -2 in the composite cobordism, with limit the rotation by π on $\mathbb{R}P^3$, term $\frac{1}{8}$ in (2.1). With index 0 and $b^+ = 0$ the levels differ by exactly $\frac{1}{8}$; computed at every such representation for five odd–odd knots in both chiralities, and for the odd–odd mirrors the minimisers are such representations.

Why the triangle cannot prove the two-step bound (proved, given DLME). The pair $-\frac{1}{n}, -\frac{1}{n+2}$ is at distance two, and its triangle has middle term $C(Y_{-1/(n+1)})^{\oplus 2}$, a closed manifold theory. By DLME's surgery lemma it is short exact with injective first map, so every class is carried into the middle term and Lemma 2.2 never yields a drop; and the first map, injective of level ≤ 0, makes the spectrum of the middle term dominate that of $Y_{-1/n}$ termwise, so $t_* = \infty$ in Proposition 4.1; the models of §3 transcribe (plausible). The bound of $\frac{1}{8}$, true in all data, lies outside the triangle, in the positions of the generators.

Census (computed). Among all 313,230 prime knots with at most 15 crossings and the 3,116 census knots with at most 10 tetrahedra, the knots with both $S^3_{\pm 2}(K)$ Seifert fibred are exactly the torus knots (11 up to mirror) and the figure-eight knot. This is rigorous (interval arithmetic, exact recognition) up to 14 crossings and for the census; for 243,282 hyperbolic 15-crossing knots it rests on floating-point cusp lengths above 12, against the threshold 6 of Agol and Lackenby. So no knot with $\Delta_K = 1$ can be tested by Seifert methods; the test must use hyperbolic surgeries, where Lemma 5.1 needs only the curve.

7. What remains, ranked

(1) *Test the mechanism on knots with $\Delta_K = 1$.* The computation of $\mathcal{C}$, Φ, the slopes and $G(Y_{\pm 2})$ for the Kinoshita–Terasaka knot, the Conway knot and the pretzel knot $P(-3,5,7)$ is now running. It decides whether Φ_0 sits at a traceless point on a branch of slope $> \frac{5}{2}$ and whether the flat connections of Y_2 avoid slopes in $(-2,0)$; if so, the drop of G is positive for these knots, given (5.1).

(2) *Prove* (5.1) *in general* (plausible, 70%): the splitting of ρ along the torus, the rule for ι, and perturbations for degenerate curves.

(3) *Control the shallow parts of $\mathcal{C}$* (conjecture; we put the probability that the drop of G is positive for every knot with $\Delta_K = 1$ at about 55%, and that this is provable along these lines at about 15%): show that Φ_0 is attained at a traceless point on a branch of slope $> \frac{5}{2}$, and that the flat connections of Y_2 with $\iota_2 = 1$ do not lie below $\Phi_0 + \frac{1}{16}$; or replace the slope hypotheses using the symmetry $(\alpha, \beta) \mapsto (\frac{1}{2} - \alpha, -\beta)$ of $\mathcal{C}$, the integrality of $\oint 2\alpha\, d\beta$, and closedness.

(4) *A drop of Λ, if wanted*: that the relevant flat connection of Y_{-2} survives in homology. A flat singular contribution supplies this at shared representations, as for the figure-eight knot.

(5) *The counting route* (Proposition 4.1; conjecture, about 35% true and under 10% provable): more classes of Y_2 than traceless bars below $\Lambda(Y_2) + \frac{1}{8}$.

(6) *Clustering forced by the twist and by a cosmetic automorphism*: it gives only $r \ge 4$ (proved for perfect complexes, if the distance-four map is an isomorphism), and nothing bounds r above; under 3%.